

Device	Sensor (What it measures)	Broker/Network (How data travels)	Subscriber (Who uses the data)	Actuator (What action it takes)
1. Sorotech Solar inverter	Solar panel Voltage, Battery voltage, charging/discharging output, power output, inverter temperature	WiFi→Home Network→sorotec monitoring platform	Homeowner, Sorotec monitoring app	Charges/discharges, battery, switches between solar/grid/battery, supplies Ac power
2. Runmefit S5 fitness tracker	Heart rate sensor, GPS, Skin temperature, accelerometer, SPO ₂	Bluetooth→Phone→Runmefit App→Runmefit Cloud	Runmefit app, user	Vibration motor display
3. Samsung Smart TV	Microphone (voice commands), ambient light sensor	WiFi → Samsung Cloud	Samsung SmartThings app, streaming services	Display, speakers
4. Smart Air Conditioner	Temperature sensor, humidity sensor	WiFi→Manufacturer Cloud	Ac app, Smartthings, Google Home, Alexa	Compressor, Fanspeed, temperature control
5. IP security camera	Camera, Microphone, motion sensor	Wifi→Cloud→App	Mobile App	Speaker, alarm, recording

Security Reflection

Device chosen: Sorotec solar Inverter

If an attacker compromised this device, what's the worst that could happen?

- **What data could they access?**
 - Solar power generation data, Electricity consumption
 - Device settings and operating status
- **What actions could they take?**
 - Turn the inverter off remotely, cutting power to the house
 - Use the inverter as an entry point to attack other devices on the home WIFI network

- Change charging/discharging settings, reducing battery life
- **How would it affect me?**
 - Privacy risk- because my energy usage patterns could reveal when I'm home or away
 - Loss of electricity or backup power